

MATH 225 – Applied Linear Algebra II

Course Notes

Obrian Buchanan

University of Waterloo

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Chapter 1: General Vector Spaces

Vector Addition Axioms

1. **Closure under Addition:** For all $u, v \in V$, $u + v \in V$.
2. **Commutativity:** $u + v = v + u$ for all $u, v \in V$.
3. **Associativity:** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$.
4. **Additive Identity:** There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
5. **Additive Inverse:** For each $v \in V$, there exists $-v \in V$ such that $v + (-v) = 0$.

Scalar Multiplication Axioms

6. **Closure under Scalar Multiplication:** For all $c \in \mathbb{R}$ and $v \in V$, $cv \in V$.
7. **Vector Distributivity:** $c(u + v) = cu + cv$ for all $c \in \mathbb{R}$ and $u, v \in V$.
8. **Scalar Distributivity:** $(c + d)v = cv + dv$ for all $c, d \in \mathbb{R}$ and $v \in V$.
9. **Associativity:** $(cd)v = c(dv)$ for all $c, d \in \mathbb{R}$ and $v \in V$.
10. **Multiplicative Identity:** $1v = v$ for all $v \in V$.

The Elements of a Vector Space

The elements of a vector space V are called **vectors**. The elements of the field $\mathbb{R}$ are called **scalars**.

Examples of Vector Spaces:

$$(1) \mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\} \text{ under standard vector addition and scalar multiplication.}$$

(2) $M_{m \times n}(\mathbb{R})$ under standard matrix addition and scalar multiplication.

(3) $P_n(\mathbb{R})$, the space of all polynomials of degree $\leq n$, defined with operations:

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (cf)(x) = cf(x) \quad (c \in \mathbb{R}).$$

Then $P_n(\mathbb{R})$ with these operations is a vector space over $\mathbb{R}$.

Fundamental Properties Derived from the Axioms

The definition of a vector space guarantees that standard arithmetic rules apply without relying on the specific nature of the objects (whether they are n -tuples, matrices, or functions).

Theorem (Elementary Arithmetic Properties of Vector Spaces): Let V be a vector space over $\mathbb{R}$. For all $\vec{x} \in V$ and all $t \in \mathbb{R}$:

1. $0\vec{x} = \vec{0}$
2. $t\vec{0} = \vec{0}$
3. $(-1)\vec{x} = -\vec{x}$
4. If $t\vec{x} = \vec{0}$, then either $t = 0$ or $\vec{x} = \vec{0}$.

Proof. **Proof of (1):** In the real numbers, $0 + 0 = 0$. By scalar distributivity:

$$0\vec{x} = (0 + 0)\vec{x} = 0\vec{x} + 0\vec{x}.$$

Since $0\vec{x} \in V$, its additive inverse $-(0\vec{x})$ exists in V . Adding this inverse to both sides:

$$\begin{aligned} 0\vec{x} + (-(0\vec{x})) &= (0\vec{x} + 0\vec{x}) + (-(0\vec{x})) \\ \vec{0} &= 0\vec{x} + (0\vec{x} + (-(0\vec{x}))) \\ \vec{0} &= 0\vec{x} + \vec{0} \\ \vec{0} &= 0\vec{x}. \end{aligned}$$

Therefore, $0\vec{x} = \vec{0}$.

Proof of (2): By the additive identity axiom, $\vec{0} + \vec{0} = \vec{0}$. Multiplying by scalar t and using vector distributivity:

$$t\vec{0} = t(\vec{0} + \vec{0}) = t\vec{0} + t\vec{0}.$$

Adding $-(t\vec{0})$ to both sides and associating terms yields $t\vec{0} = \vec{0}$.

Proof of (3): We show that $(-1)\vec{x}$ behaves as the unique additive inverse of $\vec{x}$:

$$\vec{x} + (-1)\vec{x} = 1\vec{x} + (-1)\vec{x} = (1 + (-1))\vec{x} = 0\vec{x} = \vec{0}.$$

By uniqueness of the additive inverse in V , $(-1)\vec{x} = -\vec{x}$.

Proof of (4): Suppose $t\vec{x} = \vec{0}$. If $t = 0$, the claim holds. If $t \neq 0$, the multiplicative inverse $t^{-1} \in \mathbb{R}$ exists. Multiplying both sides:

$$t^{-1}(t\vec{x}) = t^{-1}\vec{0} \implies (t^{-1}t)\vec{x} = \vec{0} \implies 1\vec{x} = \vec{0} \implies \vec{x} = \vec{0}.$$

□

Subspaces and the Subspace Test

Definition (Subspace). Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbb{R}$ under the addition and scalar multiplication operations inherited from V .

Theorem (Subspace Test). Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is a subspace of V if and only if:

1. $\vec{0} \in U$, where $\vec{0}$ is the zero vector of V (in particular, $U \neq \emptyset$).
2. For all $\vec{x}, \vec{y} \in U$, $\vec{x} + \vec{y} \in U$ (**Closure under vector addition**).
3. For all $\alpha \in \mathbb{R}$ and $\vec{x} \in U$, $\alpha\vec{x} \in U$ (**Closure under scalar multiplication**).

Proof. ($\implies$) If U is a subspace, it satisfies all vector space axioms on its own. In particular, it must be closed under addition and scalar multiplication, and contain an additive identity $\vec{0}_U \in U$.

Since $\vec{0}_U + \vec{0}_U = \vec{0}_U$, adding the inverse $-\vec{0}_U$ computed in V gives $\vec{0}_U = \vec{0}_V$.

($\impliedby$) Assume conditions (1), (2), and (3) hold. Closure under addition and scalar multiplication, as well as the existence of the additive identity, are directly satisfied.

For any $\vec{x} \in U$, setting $\alpha = -1$ ensures that $-\vec{x} = (-1)\vec{x} \in U$.

The remaining algebraic axioms (commutativity, associativity, and distributivity) hold for all elements in U automatically because they hold throughout the larger space V . Thus, U is a vector space. □

Worked Examples: Subspace Analysis

Example 1 (Polynomial with Root Constraint). Let $V = P_2(\mathbb{R})$. Define the subset:

$$U = \{p(x) \in P_2(\mathbb{R}) \mid p(3) = 0\}.$$

Prove that U is a subspace of $P_2(\mathbb{R})$.

Proof. We test the three conditions of the Subspace Test:

1. **Zero Vector:** The zero polynomial is $z(x) = 0 + 0x + 0x^2$. Evaluating at $x = 3$ gives $z(3) = 0$. Hence, $z \in U$.
2. **Closure under Addition:** Let $p(x), q(x) \in U$. Then $p(3) = 0$ and $q(3) = 0$. Consider their sum evaluated at $x = 3$:

$$(p + q)(3) = p(3) + q(3) = 0 + 0 = 0.$$

Thus, $p + q \in U$.

3. **Closure under Scalar Multiplication:** Let $p(x) \in U$ and $\alpha \in \mathbb{R}$. Then:

$$(\alpha p)(3) = \alpha \cdot p(3) = \alpha(0) = 0.$$

Thus, $\alpha p \in U$.

By the Subspace Test, U is a subspace of $P_2(\mathbb{R})$. □

Example 2 (Traceless Matrices). Let $V = M_{2 \times 2}(\mathbb{R})$. Define:

$$W = \{A \in M_{2 \times 2}(\mathbb{R}) \mid \text{tr}(A) = 0\}.$$

Recall that for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\text{tr}(A) = a + d$. Show that W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Proof. 1. **Zero Vector:** The zero matrix $O_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ satisfies $\text{tr}(O_{2 \times 2}) = 0 + 0 = 0$. Hence, $O_{2 \times 2} \in W$.

2. **Closure under Addition:** Let $A, B \in W$, so $\text{tr}(A) = 0$ and $\text{tr}(B) = 0$. By trace linearity:

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B) = 0 + 0 = 0.$$

Hence, $A + B \in W$.

3. **Closure under Scalar Multiplication:** Let $A \in W$ and $\alpha \in \mathbb{R}$. Then:

$$\text{tr}(\alpha A) = \alpha \text{tr}(A) = \alpha(0) = 0.$$

Hence, $\alpha A \in W$.

Thus, W is a subspace of $M_{2 \times 2}(\mathbb{R})$. □

Example 3 (Affine Coefficient Failure). Let $V = P_2(\mathbb{R})$ and consider the subset:

$$S = \{p(x) = ax^2 + bx + c \in P_2(\mathbb{R}) \mid a - 2b + c = 3\}.$$

Determine whether S is a subspace of V .

Solution: The zero polynomial $z(x) = 0x^2 + 0x + 0$ has coefficients $a = 0, b = 0, c = 0$. Evaluating the defining equation:

$$a - 2b + c = 0 - 2(0) + 0 = 0 \neq 3.$$

Because the zero polynomial is not in S , condition (1) of the Subspace Test fails immediately. Thus, S is **not** a subspace of $P_2(\mathbb{R})$.

Linear Span and Minimal Subspaces